THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 225.5 m²	Canvas = CSA of cylinder + CSA of cone; $= 88 + 137.5 = 225.5$ m²
2	**(a)** 32.4	$\sum f_i = 50$ and $\sum f_i x_i = 1620$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1620}{50} = 32.4$
3	**(c)** $\frac{1}{4}$	Favourable outcomes: (1,3), (2,2), (2,6), (3,1), (3,5), (4,4), (5,3), (6,2), (6,6); number $= 9$; $P = \frac{9}{36} = \frac{1}{4}$
4	**(b)** 396 cm³	$\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(6^3 - 3^3)$; Volume $= 396$ cm³
5	**(c)** 26	Median class 20–30: $l = 20$, $cf = 16$, $f = 15$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 20 + \frac{25-16}{15} \times 10 = 26$
6	**(d)** $\frac{7}{8}$	Favourable outcomes: HHT, HTH, HTT, THH, THT, TTH, TTT; number $= 7$; $P = \frac{7}{8}$
7	**(c)** 6835.5 cm³	Cube $= 9261$ cm³; hemisphere of radius 10.5 cm $= \frac{2}{3}\pi r^3 = 2425.5$ cm³; Volume $= 6835.5$ cm³
8	**(c)** 77	$l = 70$, $f_1 = 16$, $f_0 = 2$, $f_2 = 10$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 70 + \frac{14}{20} \times 10 = 77$
9	**(a)** $\frac{7}{15}$	Favourable numbers: 3, 5, 6, 9, 10, 12, 15, 18, 20, 21, 24, 25, 27, 30; number $= 14$; $P = \frac{14}{30} = \frac{7}{15}$
10	**(c)** 256 cm	$\frac{4}{3}\pi(12)^3 = \pi(3)^2 L$; $L = 256$ cm
11	**(c)** 6	$\frac{1750 + 35p}{44 + p} = 39.2$ gives $(35 - 39.2)p = -25.2$; $p = 6$
12	**(c)** 6	$\frac{x}{3+x} = 2 \times \frac{3}{3+x}$ gives $x = 2 \times 3$; $x = 6$
13	**(c)** 1 cm	$\frac{4}{3}\pi(3)^3 = \pi(6)^2 x$; $x = 1$ cm
14	**(a)** 11	$91.5 = 90 + \frac{50 - (36 + x)}{20} \times 10$ gives $x = 11$; $x = 11$; then $y = 100 - 68 - x = 21$
15	**(a)** 2	Cross-multiplying: $2(3 + x) = 1(8 + x)$; $x = 2$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Volume of sphere = volume of cone; $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(6)^2(3)$ gives $R^3 = 27$ (1); $R = 3$ cm (1).	1+1
17	Take $a = 35$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2; $\sum f_i = 50$ and $\sum f_i u_i = -13$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 35 + 10 \times \frac{-13}{50} = 32.4$ (1).	1+1
18	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,2), (1,5), (2,1), (2,4), (3,3), (3,6), (4,2), (4,5), (5,1), (5,4), (6,3), (6,6); number $= 12$ (1); $P = \frac{12}{36} = \frac{1}{3}$ (1).	1+1
19	Volume of cylinder $= \pi \times 6^2 \times 48$; Volume of a sphere $= \frac{4}{3}\pi(3)^3$; $n = \frac{3r^2 h}{4r^3}$ (1); $n = 48$ (1).	1+1
20	Cumulative frequencies: 4, 8, 18, 26, 34, 40; $N = 40$, so $\frac{N}{2} = 20$; Median class 50–60: $l = 50$, $cf = 18$, $f = 8$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 50 + \frac{20-18}{8} \times 10 = 52.5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 4158$ cm³; two hemispheres $= 4851$ cm³ (1); Volume $= 9009$ cm³ (1).	1+1+1
22	Take $a = 35$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2 (1); $\sum f_i = 50$ and $\sum f_i u_i = -48$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 35 + 10 \times \frac{-48}{50} = 25.4$ (1).	1+1+1
23	$P(\text{blue}) = \frac{x}{7+x}$ and $P(\text{red}) = \frac{7}{7+x}$ (1); $\frac{x}{7+x} = 2 \times \frac{7}{7+x}$ gives $x = 2 \times 7$ (1); $x = 14$ (1).	1+1+1
24	Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 19404$ cm³; cone $= \frac{1}{3}\pi r^2 h = 6468$ cm³ (1); Volume $= 25872$ cm³ (1).	1+1+1
	OR: Cumulative frequencies: 6, 15, 22, 27, 43, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 30–40: $l = 30$, $cf = 22$, $f = 5$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 30 + \frac{25-22}{5} \times 10 = 36$ (1).	1+1+1

25	$\frac{3+x}{9+x} = \frac{1}{2}$ (1); Cross-multiplying: $2(3+x) = 1(9+x)$ (1); $x = 3$ (1).	1+1+1
	OR: Surface = CSA of cylinder + two hemispherical hollows (the flat ends are removed) (1); $2\pi rh + 2(2\pi r^2)$ with $r = 3.5$, $h = 15$ (1); TSA $= 330 + 154 = 484$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(6)^2(20) = \pi(4)^2 x$ (2); $x = 15$ cm (1).	2+2+1
	OR: $\sum f_i = 45 + p$ and $\sum f_i x_i = 1445 + 35p$ (2); $\frac{1445+35p}{45+p} = 32.4$ gives $(35 - 32.4)p = 13$ (2); $p = 5$ (1).	2+2+1
27	Total outcomes $= 36$ (2); Favourable outcomes: 4; 15; 12 (2); Probabilities: $\frac{1}{9}$, $\frac{5}{12}$, $\frac{1}{3}$ (1).	2+2+1
	OR: Volume of water is the same in both tanks (2); $\pi r^2 h = 3080$ cm^3 and base area of cuboid $= 308$ cm^2 (2); Height $= 3080 \div 308 = 10$ cm (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 1056 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 7 \times 10 + 4 \times \frac{22}{7} \times 7^2$; TSA $= 1056$ cm^2
2	**(b)** 22	$35 = 30 + \frac{50-(24+x)}{10} \times 10$ gives $x = 21$; $x = 21$; then $y = 100 - 57 - x = 22$
3	**(a)** $\frac{1}{4}$	Favourable outcomes: (2,2), (2,4), (2,6), (4,2), (4,4), (4,6), (6,2), (6,4), (6,6); number $= 9$; $P = \frac{9}{36} = \frac{1}{4}$
4	**(b)** 20328 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 42 \times 70 + 2 \times \frac{22}{7} \times 42^2 = 20328$ cm^2
5	**(a)** 30	$\frac{1200}{40}$
6	**(a)** $\frac{5}{11}$	Favourable letters: E, U, A, I, O; number $= 5$; $P = \frac{5}{11}$
7	**(d)** 192	$n = \frac{24 \times 12 \times 18}{3^3}$; $n = 192$
8	**(d)** 112	$\sum f_i = 50$ and $\sum f_i x_i = 5600$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{5600}{50} = 112$
9	**(c)** $\frac{5}{12}$	Favourable outcomes: 15 pairs; number $= 15$; $P = \frac{15}{36} = \frac{5}{12}$
10	**(b)** 357.5 m²	Canvas = CSA of cylinder + CSA of cone; $= 220 + 137.5 = 357.5$ m^2
11	**(a)** 38	Median class 30–40: $l = 30$, $cf = 21$, $f = 5$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 30 + \frac{25-21}{5} \times 10 = 38$
12	**(b)** $\frac{1}{2}$	Favourable outcomes: HHH, HHT, HTH, HTT; number $= 4$; $P = \frac{4}{8} = \frac{1}{2}$
13	**(b)** 1056 cm³	$\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(8^3 - 2^3)$; Volume $= 1056$ cm^3
14	**(a)** 85	$l = 80$, $f_1 = 13$, $f_0 = 5$, $f_2 = 5$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 80 + \frac{8}{16} \times 10 = 85$
15	**(b)** $\frac{27}{40}$	Favourable numbers: 27 outcomes; number $= 27$; $P = \frac{27}{40}$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Total possible outcomes $= 36$; Favourable numbers: 6, 12, 18, 24, 30, 36; number $= 6$ (1); $P = \frac{6}{36} = \frac{1}{6}$ (1).	1+1
17	TSA = surface of cube − circle covered + curved surface of hemisphere; $6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$ with $r = 17.5$ cm (1); TSA $= 7350 + 962.5 = 8312.5$ cm^2 (1).	1+1
18	Highest frequency is 16, so the modal class is 70–80; $l = 70$, $f_1 = 16$, $f_0 = 2$, $f_2 = 10$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 70 + \frac{14}{20} \times 10 = 77$ (1).	1+1
19	Total balls $= 6 + 4 + 5 = 15$; Favourable balls $= 10$ (1); $P = \frac{10}{15} = \frac{2}{3}$ (1).	1+1
20	Volume of sphere = volume of cone; $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(3)^2(96)$ gives $R^3 = 216$ (1); $R = 6$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$\sum f_i = 50$ and $\sum f_i x_i = 3360$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{3360}{50} = 67.2$ (1); $l = 70$, $f_1 = 19$, $f_0 = 10$, $f_2 = 3$, $h = 10$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 70 + \frac{9}{25} \times 10 = 73.6$ (1).	1+1+1
22	Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (2,6), (3,5), (4,4), (5,3), (6,2); number $= 5$ (1); $P = \frac{5}{36}$ (1).	1+1+1
23	Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 6237$ cm^3; two hemispheres $= 4851$ cm^3 (1); Volume $= 11088$ cm^3 (1).	1+1+1
24	Take $a = 45$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2 (1); $\sum f_i = 50$ and $\sum f_i u_i = -44$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 45 + 10 \times \frac{-44}{50} = 36.2$ (1).	1+1+1
	OR: $P(\text{blue}) = \frac{x}{8+x}$ and $P(\text{red}) = \frac{8}{8+x}$ (1); $\frac{x}{8+x} = 3 \times \frac{8}{8+x}$ gives $x = 3 \times 8$ (1); $x = 24$ (1).	1+1+1
25	Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 19404$ cm^3; cone $= \frac{1}{3}\pi r^2 h = 19404$ cm^3 (1); Volume $= 38808$ cm^3 (1).	1+1+1

	OR: Cumulative frequencies: 3, 9, 22, 37, 44, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 110–120: $l = 110$, $cf = 22$, $f = 15$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 110 + \frac{25-22}{15} \times 10 = 112$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Median class is 50–60 with $f = 20$, and the cumulative frequency before it is $cf = 21 + x$ (2); $54 = 50 + \frac{50-(21+x)}{20} \times 10$ gives $x = 21$ (2); $x = 21$; then $y = 100 - 63 - x = 16$ (1).	2+2+1
	OR: Total outcomes $= 40$ (2); Favourable counts: 13, 12, 10 (2); Probabilities: $\frac{13}{40}$, $\frac{3}{10}$, $\frac{1}{4}$ (1).	2+2+1
27	Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(6)^2(16) = \pi(4)^2 x$ (2); $x = 12$ cm (1).	2+2+1
	OR: $\sum f_i = 37 + p$ and $\sum f_i x_i = 4235 + 105p$ (2); $\frac{4235+105p}{37+p} = 112$ gives $(105 - 112)p = -91$ (2); $p = 13$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 54684 cm³	Cube $= 74088$ cm^3; hemisphere of radius 21 cm $= \frac{2}{3}\pi r^3 = 19404$ cm^3; Volume $= 54684$ cm^3
2	**(c)** 5	$\frac{1635+45p}{45+p} = 37.2$ gives $(45-37.2)p = 39$; $p = 5$
3	**(a)** 12	$\frac{x}{4+x} = 3 \times \frac{4}{4+x}$ gives $x = 3 \times 4$; $x = 12$
4	**(d)** 243 cm	$\frac{4}{3}\pi(9)^3 = \pi(2)^2 L$; $L = 243$ cm
5	**(c)** 10	$45 = 40 + \frac{25-(7+x)}{16} \times 10$ gives $x = 10$; $x = 10$; then $y = 50 - 30 - x = 10$
6	**(a)** 6	Cross-multiplying: $8(4+x) = 5(10+x)$; $x = 6$
7	**(c)** 2 cm	$\frac{4}{3}\pi(6)^3 = \pi(12)^2 x$; $x = 2$ cm
8	**(c)** 16	$54 = 50 + \frac{50-(21+x)}{20} \times 10$ gives $x = 21$; $x = 21$; then $y = 100 - 63 - x = 16$
9	**(c)** $\frac{11}{36}$	Favourable outcomes: (1,6), (2,6), (3,6), (4,6), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6); number $= 11$; $P = \frac{11}{36}$
10	**(b)** 3344 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 14 \times 10 + 4 \times \frac{22}{7} \times 14^2$; TSA $= 3344$ cm^2
11	**(a)** just greater than $\frac{N}{2}$	Median class definition
12	**(b)** $\frac{4}{11}$	Favourable letters: O, A, I, I; number $= 4$; $P = \frac{4}{11}$
13	**(d)** 13728 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 28 \times 100 + 2 \times \frac{22}{7} \times 28^2 = 13728$ cm^2
14	**(a)** 103	$\sum f_i = 100$ and $\sum f_i x_i = 10300$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{10300}{100} = 103$
15	**(a)** $\frac{1}{36}$	Favourable outcomes: (6,6); number $= 1$; $P = \frac{1}{36}$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Take $a = 115$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2; $\sum f_i = 50$ and $\sum f_i u_i = -15$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 115 + 10 \times \frac{-15}{50} = 112$ (1).	1+1
17	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,2), (2,1), (2,4), (3,6), (4,2), (6,3); number $= 6$ (1); $P = \frac{6}{36} = \frac{1}{6}$ (1).	1+1
18	Volume of cylinder $= \pi \times 6^2 \times 72$; Volume of a sphere $= \frac{4}{3}\pi(2)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 243$ (1).	1+1
19	Cumulative frequencies: 8, 10, 13, 33, 36, 39, 50; $N = 50$, so $\frac{N}{2} = 25$; Median class 40–50: $l = 40$, $cf = 13$, $f = 20$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 40 + \frac{25-13}{20} \times 10 = 46$ (1).	1+1
20	Total possible outcomes $= 42$; Favourable numbers: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41; number $= 13$ (1); $P = \frac{13}{42}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$\frac{5+x}{10+x} = \frac{2}{3}$ (1); Cross-multiplying: $3(5+x) = 2(10+x)$ (1); $x = 5$ (1).	1+1+1
22	Surface = CSA of cylinder + two hemispherical hollows (the flat ends are removed) (1); $2\pi rh + 2(2\pi r^2)$ with $r = 7$, $h = 10$ (1); TSA $= 440 + 616 = 1056$ cm^2 (1).	1+1+1
23	$\sum f_i = 50$ and $\sum f_i x_i = 1960$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1960}{50} = 39.2$ (1); $l = 50$, $f_1 = 14$, $f_0 = 8$, $f_2 = 5$, $h = 10$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 50 + \frac{6}{15} \times 10 = 54$ (1).	1+1+1
24	Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1); number $= 6$ (1); $P = \frac{6}{36} = \frac{1}{6}$ (1).	1+1+1
	OR: Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 16632$ cm^3; two hemispheres $= 38808$ cm^3 (1); Volume $= 55440$ cm^3 (1).	1+1+1
25	Take $a = 45$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2 (1); $\sum f_i = 40$ and $\sum f_i u_i = -7$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 45 + 10 \times \frac{-7}{40} = 43.25$ (1).	1+1+1

	OR: $P(\text{blue}) = \frac{x}{9+x}$ and $P(\text{red}) = \frac{9}{9+x}$ (1); $\frac{x}{9+x} = 2 \times \frac{9}{9+x}$ gives $x = 2 \times 9$ (1); $x = 18$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Total outcomes $= 36$ (2); Favourable outcomes: 6; 14; 7 (2); Probabilities: $\frac{1}{6}$, $\frac{7}{18}$, $\frac{7}{36}$ (1).	2+2+1
	OR: Volume of water is the same in both tanks (2); $\pi r^2 h = 4620$ cm^3 and base area of cuboid $= 231$ cm^2 (2); Height $= 4620 \div 231 = 20$ cm (1).	2+2+1
27	Median class is 120–130 with $f = 20$, and the cumulative frequency before it is $cf = 27 + x$ (2); $120.5 = 120 + \frac{50-(27+x)}{20} \times 10$ gives $x = 22$ (2); $x = 22$; then $y = 100 - 57 - x = 21$ (1).	2+2+1
	OR: Total outcomes $= 50$ (2); Favourable counts: 7, 10, 1 (2); Probabilities: $\frac{7}{50}$, $\frac{1}{5}$, $\frac{1}{50}$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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